ADDING ONE NONDEGENERATE SIMPLEX GIVES A COVERING ON HOM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1828-1851 of sources/stacks-hypercovering/pinned-original-source.tex; proof span is lines 1853-1887, the complete author proof (33 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 17 of queue_hypercovering.json (pb=1184, stmt 1828-1851, proof 1853-1887).

Editorial note. stacks-hypercovering wave-1 (hc01): fresh adjudication accept (2026-10-14); pool per standing fallback order (hypercovering -> pic -> spaces-more-cohomology) after md01 closed the moduli pool; hypercovering.tex 53 labels/34 with proof judged 26 accept / 8 reject / 14 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 01G4: sources/stacks-hypercovering/pinned-original-source.tex, source lines 524–535. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 018V: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1887–1887. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 018R: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1887–1887. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 017N: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1887–1887. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:hypercovering.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0177: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1887–1887. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{C}$ be a site with fibre products. Let X be an object of $\mathcal{C}$. Let K be a hypercovering of X. Let $U \subset V$ be simplicial sets, with U_n, V_n finite nonempty for all n. Assume that U has finitely many nondegenerate simplices. Suppose $n \geq 0$ and $x \in V_n$, $x \notin U_n$ are such that*

(1) *$V_i = U_i$ for $i < n$,*
(2) *$V_n = U_n \cup \{x\}$,*
(3) *any $z \in V_j$, $z \notin U_j$ for $j > n$ is degenerate.*

Then the morphism

$$\mathrm{Hom}(V, K)_0 \longrightarrow \mathrm{Hom}(U, K)_0$$

of $SR(\mathcal{C}, X)$ is a covering.

Proof. If $n = 0$, then it follows easily that $V = U \amalg \Delta[0]$ (see below). In this case $\mathrm{Hom}(V, K)_0 = \mathrm{Hom}(U, K)_0 \times K_0$. The result, in this case, then follows from Lemma 01G4.

Let $a : \Delta[n] \to V$ be the morphism associated to x as in Simplicial, Lemma 0177. Let us write $\partial\Delta[n] = i_{(n-1)!}\mathrm{sk}_{n-1}\Delta[n]$ for the $(n-1)$-skeleton of $\Delta[n]$. Let $b : \partial\Delta[n] \to U$ be the restriction of a to the $(n-1)$ skeleton of $\Delta[n]$. By Simplicial, Lemma 018R we have $V = U \amalg_{\partial\Delta[n]} \Delta[n]$. By Simplicial, Lemma 017N we get that

$$\begin{array}{ccc} \mathrm{Hom}(V, K)_0 & \longrightarrow & \mathrm{Hom}(U, K)_0 \\ \downarrow & & \downarrow \\ \mathrm{Hom}(\Delta[n], K)_0 & \longrightarrow & \mathrm{Hom}(\partial\Delta[n], K)_0 \end{array}$$

is a fibre product square. Thus it suffices to show that the bottom horizontal arrow is a covering. By Simplicial, Lemma 018V this arrow is identified with

$$K_n \to (\mathrm{cosk}_{n-1}\mathrm{sk}_{n-1}K)_n$$

and hence is a covering by definition of a hypercovering. □